Decoding An ELECTRIC VEHICLE

In an official document sent out by Ministry of Power on December 14, 2018, the government of India has outlined several key facts, figures and a plan to support the expansion of EVs in the country

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Electric vehicles, also called EVs, use one or more electric or traction motors for propulsion. These may be powered through a collector system using electricity from off-vehicle sources, or may be self-contained with a battery, solar panels or an electric generator to convert fuel into electricity. EVs include, but are not limited to, road and rail vehicles, surface and underwater vessels, electric aircraft and electric spacecraft.

EVs first came into existence in the mid-19th century, when electricity was among the preferred methods for motor vehicle propulsion, providing a level of comfort and ease of operation that could not be achieved by petrol cars of the time. Limitations of heavy weight, short trip range, long charging time and poor durability of batteries compared to later internal combustion engine vehicles led to a worldwide decline in their use; although, EVs have continued to be used in the form of electric trains and other niche uses.

At the beginning of the 21st century, interest in electric and other alternative fuel vehicles increased due to growing concerns over the problems associated with hydrocarbon-fueled vehicles, including damage to environment caused by their emissions, sustainability of the current hydrocarbon-based transportation infrastructure as well as improvements in EV technology.

Some advantages of EVs are:

- Most electric motors can travel 150km to 180km before these need to be charged.
- No tail pipe exhaust means no greenhouse gases such as carbon-dioxide, NOx pollution and PM10.
- No oil consumption means less reliance on fossil fuel.
- Cars can be recharged whenever convenient to user.
- More cost-effective than regular cars because of long-lasting battery use.
- Cheaper to maintain because of fewer moving parts.
- Creates less noise pollution because of a silent engine.

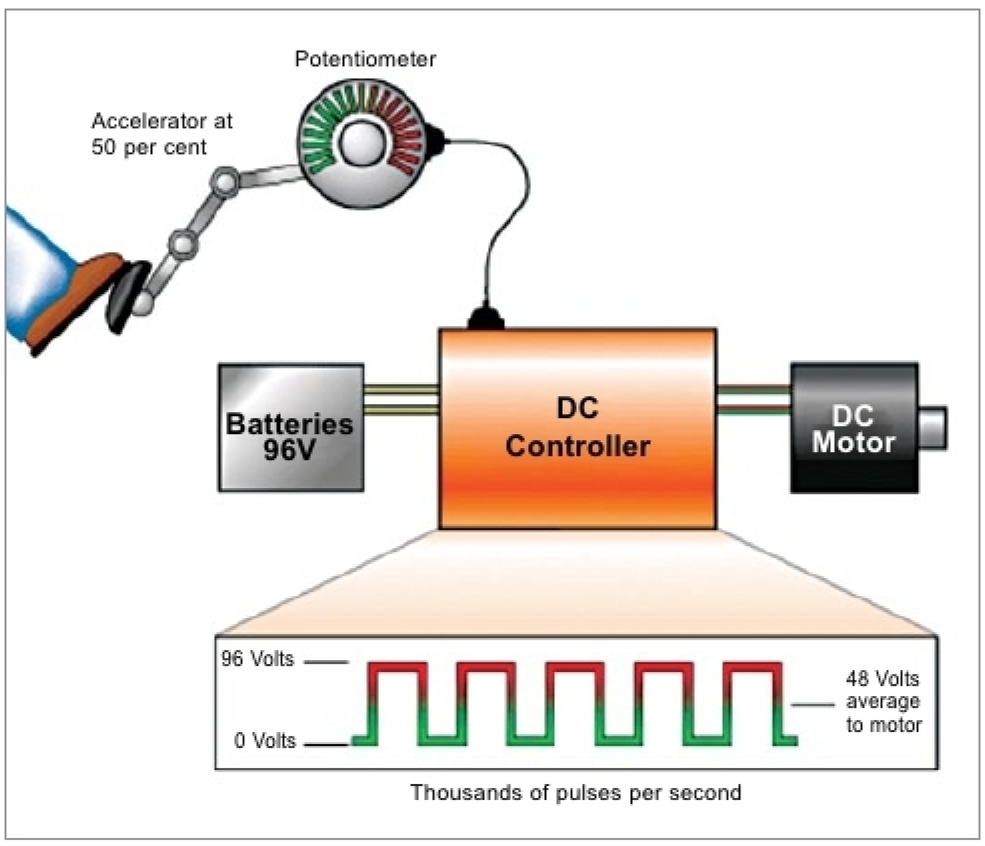

Fig. 1: Components of an EV